# Class 10
# PHYSICS
# Solutions

## Table of Content

1. Current Electricity 1 – 29

# 1. CURRENT ELECTRICITY_KEY & SOLUTIONS

## WORKSHEET-1

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. A | 2. B | 3. B | 4. A | 5. A | 6. A | 7. A |
| 8. D | 9. B | 10. C | 11. A | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. B | 2. C | 3. B | 4. A | 5. A | 6. B | 7. D |
| 8. C | 9. C | 10. B | 11. D | 12. B | 13. C | 14. C |
| 15. B | 16. B | 17. D | 18. B | 19. A | 20. B | |

### Level – III

| | | | | | |
|---|---|---|---|---|---|
| 21. B | 22. A | 23. A | 24. A | 25. A | |
| 26. a-p,t; b-p,s; c-q,r; d-q | 27. A | 28. A | 29. B | 30. D | |

## SOLUTIONS

### Level – II

1. $i = \frac{ne}{t} = \frac{6\times10^{17}\times1.6\times10^{19}}{60} = 1.6$ mA
2. $i = \frac{q}{t} = \frac{900}{10\times60} = \frac{3}{2} = 1.5$ A
3. $i = \frac{q}{t} = \frac{ne}{t}$

   $4 = \frac{n}{t} \times 1.6 \times 10^{-19}$

   $\frac{n}{t} = \frac{4}{1.6\times10^{-19}} = 2.5 \times 10^{19}$
4. $i = 2\mu A = 2 \times 10^{-6}$ A

   $t = 5$ minutes $= 5 \times 60$sec

   $Q = ?$

   $i = \frac{Q}{t}$

   $Q = it = 2 \times 10^{-6} \times 5 \times 60$

   $Q = 6 \times 10^{-4}$C.
5. $V_d = \frac{eE}{m}\tau = \frac{e}{m}\left[\frac{v}{l}\right]\tau$;

   $V_d \alpha v$
6. $\frac{(n_e+n_p)e}{t}$
7. $i = 1$ A

   $A = 1$ mm$^2 = 1 \times 10^{-6}$ m$^2$

   Atomic weight (A) $= 63.5$amu $= 63..5 \times 10^{-3}$Kg

   Avogadro number (N) $= 6.023 \times 10^{23}$

   Density of metal (d) $= 900$Kgm$^{-3}$

   Number of electrons ( no) $= 1$

   free electron density $(n) = \frac{N\,d}{A}$

   $n = \frac{6.023\times10^{23}\times1\times9000}{63.5\times10^{-3}}$

   $= 0.853 \times 10^{29}$

$$V_d = \frac{i}{n_e A} = \frac{1}{0.853 \times 10^{29} \times 1.6 \times 10^{-19} \times 1 \times 10^{-6}}$$
$$V_d = 0.074 \text{mms}^{-1}$$

8. $i = AnqV_d$
$i \alpha r^2 v \Rightarrow \frac{V_A}{V_B} = \frac{i_A}{i_B}\left[\frac{r_A}{r_B}\right]^2$
$$\frac{1}{1.2} \cdot \frac{9}{4} = \frac{10}{12} \times \frac{9}{4} = \frac{15}{8}$$

9. $i = Anv_d q$
$i \alpha AV$
$$\frac{V_2}{V_1} = \frac{i_2}{i_1} \cdot \frac{A_1}{A_2} = \frac{2i}{i} - \frac{1}{2} = 1$$
$V_2 = V_1 = V$

10. $V_d = \left(\frac{e\tau}{m}\right) E$

11. Free electron density (n) $= 8.5 \times 10^{28}$ m$^3$
$i = 1$ A
$A = 1$ cm$^2$ $= 1 \times 10^{-4}$ m$^2$
$$V_d = \frac{i}{neA}$$
$$= \frac{1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-14}}$$
$$= 0.0735 \times 10^{-5} \text{ m/s}$$
$V_d = 0.735 \times 10^{-6}$ m/s $= 0.735 \mu$m/s
$$\tau = \frac{l}{V_d} = \frac{10^4}{0.735 \times 10^{-6}}$$
$$\tau = \frac{1.3605 \times 10^{10}}{60 \times 60 \times 24 \times 365}$$
$= 43.1$ yrs.

12. $V_{d_1} = v, v_{d_2} = ?$
$r_1 = r, r_2 = 2r; i_1 = i, i_2 = 2i$
$$\frac{V_{d_1}}{v_{d_2}} = \frac{i}{2i} \times \frac{r_2^2}{r_1^2}$$
$$\frac{V}{v_{d_2}} = \frac{i}{2i} \times \left[\frac{2r}{r}\right]^2$$
$$v_{d_2} = \frac{V}{2}$$

13. $q = \int_{t_1}^{t_2} i dt$
$I = 3t + 4t^2$
$$\frac{dq}{dt} = (3t + 4t^2)$$
$$dq = (3t + 4t^2)dt$$
$$\int dq = \int_1^3 (3t + 4t^2)dt$$
$$= \frac{3 \cdot t^2}{2} + \frac{4 \cdot t^3}{3} = \frac{3}{2}(3^2 - 12) + \frac{4}{3}(3^3 - 1^3)$$
$$= \frac{3}{2}(8) + \frac{4}{3}(28) = 12 + \frac{112}{3}$$
$$= \frac{36 + 112}{3} = \frac{140}{3}$$

14. $V_d \alpha \frac{1}{A}$

15. $V_d = \frac{i}{neA}$

$$v_d = \frac{i}{neA}$$

$$v_d = \frac{16}{4 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-5}} = \frac{16}{4 \times 16 \times 10}$$

$$= \frac{1}{4} \times 10^{-3} = \frac{10}{4} \times 10^{-4}$$

$v_d = 2.5 \times 10^{-4}$ m/s

16. i = 3.6 A

t = 3 h = $3 \times 60 \times 60$ s

Q = ?

Q = it

$Q = 3.6 \times 3 \times 60 \times 60$

$Q = 3.9 \times 10^4$c

17. n = 1

e = $-1.6 \times 10^{-19}$c

v = $2.2 \times 10^6$ m/s

r = $5.3 \times 10^{-11}$ m

$$f = \frac{v}{2\pi r} = \frac{2.2 \times 10^6}{2 \times 3.142 \times 5.3 \times 10^{-11}}$$

$= 0.066 \times 10^{17}$

f = 6.6rps

$i = \frac{-neV}{2\pi r}$ (or) i = −nef

$i = (-1) \times (-1.6 \times 10^{-19}) \times (6.6)$

$i = 1.06 \times 10^{-3}$ A

i = 1.06 mA

18. $v_d = \frac{i}{neA}$

$$= \frac{4}{10^{28} \times 1.6 \times 10^{-19} \times 10^{-8}}$$

$$= \frac{4}{16} = \frac{1}{4} = 0.25$$

19.

$$j = \frac{i}{A} = \frac{3.6}{(6 \times 10^{-3})^2} = \frac{3.6}{36 \times 10^{-6} \times 10} = 1 \times 10^5 \text{ A/m}^2$$

20. $V_d = \frac{\tau}{ne} = \frac{I}{Ane} = \frac{1}{(2\times10^{-6})^2\times(8.5\times10^{22}\times10^6)(1.6\times10^{-19})}$

= 0.036 mm/s

**Level – III**

21. Conceptual
22. Conceptual
23. $J = \frac{I}{A} = \frac{1.8}{0.5\times10^{-6}} = 3.6 \times 10^6 Am^{-2}$
24. 1hr15 mints = 4500 s.

$$i = \frac{q}{t} = \frac{9}{4500} = 0.02$$

$$j = \frac{i}{A} = \frac{0.02 \times 4}{3.14 \times 1 \times 10^{-6}}$$

$$V_d = \frac{j}{nE} = \frac{0.02 \times 4}{3.14 \times 10^{-6} \times 5.8 \times 10^{28} \times 1.6 \times 10^{-6}} = 2.72 \times 10^{-6} \text{ A/m}^2$$

25. $r_1 = 2$ cm, $r_2 = 4$ cm.

ratio of current density $= \frac{j_1}{j_2} = \frac{r_2^2}{r_1^2} = \frac{4^2}{2^2} = \frac{4}{1}$

26. Conceptual
27. $\tau = \frac{i}{A}$

28. $i = \frac{dq}{dt} = \frac{d(2t-6t^2+10t^3)}{dt} = (2 - 12t + 30t^2)$
initially $t = 0$,
$i = 2$ A

29. current is minimum $\frac{di}{dt} = 0$
$= -12 + 60t = 0$
$t = \frac{1}{5}$ s

30. min imum value of current $i = 2 - 12 \times \frac{1}{5} + 30 \times \left(\frac{1}{5}\right)^2 = 0.8$ A

## WORKSHEET-2

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. A | 2. B | 3. C | 4. C | 5. D | 6. C | 7. A |
| 8. D | 9. B | 10. C | 11. B | 12. A | 13. A | 14. D |
| 15. A | 16. D | 17. C | 18. B | 19. D | 20. B | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. D | 2. D | 3. A | 4. A | 5. B | 6. B | 7. B |
| 8. B | 9. D | 10. C | 11. D | 12. A | 13. B | 14. D |
| 15. C | 16. A | 17. D | 18. C | 19. B | 20. A | 21. C |

### Level – III

| | | | | |
|---|---|---|---|---|
| 22. A, B, C, D | | 23. A | 24. A | 25. B | 26. A |
| 27.a-q,t; b-r,s; c-q,t; d-p | | 28. C | 29. C, D | 30. a-p,t; b-q,t; c-s,t; d-r,t |
| 31. C | 32. A | 33. A | | |

## SOLUTIONS

### Level – II

1. $Ral^2$
$\frac{R_1}{R_2} = \left(\frac{1_1}{1_2}\right)^2$
$\frac{R_1}{R_2} = \left(\frac{1}{21}\right)^2 \Rightarrow R_2 = 4R_1$

2. $R = \frac{1}{\pi(R_2{}^2 - R_1{}^2)} = \frac{1.7\times10^{-5}\times5}{3.14((0.05)^2-(0.045)^2)} = 0.057\Omega m$

3. $J = \frac{RA}{1} = \frac{24\times\pi(0.25\times10^{-3})^2}{4} = 1.1775 \times 10^{-6}\Omega m$

4. $R\alpha 1^2 \Rightarrow \frac{R_1}{R_2} = \left(\frac{1_1}{1_2}\right)^2 \Rightarrow \frac{100}{R_2} = \left(\frac{1}{21}\right)^2 \Rightarrow \frac{100}{R_2} = \frac{1}{4} \uparrow R$
$R_2 = 400\Omega$

5. $R = \frac{\rho l}{A}$
$1 = \frac{RA}{\rho} = \frac{1000 \times 0.01 \times 10^{-6}}{1.7 \times 10^{-8}} = 0.588$ km
$= 0.6$ km

6. $R = \frac{s}{10}$

7. $R_1 = \frac{1}{10^{-6}} = 10^6$

$$R_2 = \frac{0.5}{2\times10^{-6}} = 20\times10^6$$

$$R_3 = \frac{2}{10^{-6}} = 2\times10^6$$

$$R_4 = \frac{4}{2\times10^{-6}} = 2\times10^6$$

8. $\frac{R_1}{R_2} = \frac{1_1 1_1}{A_1}\times\frac{A_2}{1_2 1_2} = \left(\frac{3}{4_2}\right)\left(\frac{2}{9}\right)\left(\frac{27^3}{8}\right) = \frac{9}{16}$

9. $\frac{R_1}{R_2} = \frac{1_1}{A_1}\times\frac{A_2}{1_2} = \frac{11}{12}\left(\frac{8}{3}\right) = \frac{8}{6} = 4{:}3.$

$$\left[\frac{3}{4} = \frac{m_1}{m_2} = \frac{A_1 1_1}{A_2 1_2}\right.$$

$$\left.\frac{3}{16} = \frac{A_1}{A_2}\left(\frac{1}{2}\right)\right]$$

10. $R^1 = n^4 R = (2^4)R$

$= 16(\mathrm{R})$

$= 16(18)$

11. $\frac{R_1}{R_2} = \frac{l_1}{A_1}\left(\frac{A_2}{1_2}\right) = \frac{1_1}{1_2^2} = \frac{1}{4}$

$$\left[\mathrm{A}_1 1_1 = \mathrm{A}_2 1_2\right.$$

$$\left.\frac{1_1}{1_2} = \frac{\mathrm{A}_2}{\mathrm{A}_1}\right]$$

12. $\frac{R_1}{R_2} = \frac{1}{4}, \frac{1_1}{1_2} = \frac{1}{2}$

$$\frac{1_1}{A_1} = \frac{A_2}{1_2} = \frac{1}{4} = \frac{1_1}{1_2}\left(\frac{A_2}{A_1}\right) \Rightarrow \frac{1}{4} = \frac{1}{2}\left(\frac{A_2}{A_1}\right)$$

$$\frac{A_1}{A_2} = 2;\ \frac{m_1}{m_2} = \frac{A_1 1_1}{A_2 1_2}$$

$$= 2\left(\frac{1}{2}\right)$$

13. $\frac{R_1}{R_2} = \frac{6}{0.25}\frac{(2.25)^9}{10^5} = \frac{27}{5}$

14. $R_1 : R_2 : R_3 = l_1\frac{1_1}{A_1} : l_2\frac{l_2}{A_2} : l_3\frac{1_3}{A_3}$

$$\therefore R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3} = \frac{25}{1} : \frac{9^3}{3} : \frac{1}{5}$$

$$= 125 : 15 : 1$$

15. $10 = \frac{l}{A}\frac{R}{10} = \frac{2l}{\frac{\gamma^2}{4}}\times\frac{\gamma^2}{l} = 8 \Rightarrow R = 80$

16. $\rho = \frac{RA}{l} = \frac{26\times\pi\times0.825\times10^{-6}}{1} = 1.84\times10^{-6}$

17. $\frac{R_2}{R_1} = \frac{l_2}{A_2}\frac{A_1}{l_1} = \frac{l_2}{l_1}\left(\frac{l_2}{l_1}\right)$

$$R_2 = 4R_1$$

$$\frac{l_1}{l_2} = \frac{A_2}{A_1};\ \frac{l_1}{l_2} = \frac{10}{20}$$

18.. $l_1\pi\gamma_1^2 = l_2\pi\gamma_2^2$

$$\frac{l_1}{l_2} = \frac{\gamma_2^2}{\gamma_1^2}; \frac{l_1}{l_2} = 4$$

$$R_2 = \left(\frac{l_2}{\gamma_2^2}\frac{\gamma_1^2}{l_1}\right)R_1$$

$$= \frac{1}{4}\left(\frac{1}{4}\right)4 = 0.25\Omega$$

19. $R^1 = n^4 R$

$= 4^4(10) = 2560$

20. $2 = \frac{A_2}{A_1}R_2 = \frac{l_2}{A_2}\left(\frac{A_1}{l_1}\right)R_1 \Rightarrow R_2 = \frac{1}{2}\left(\frac{1}{2}\right)R$

$\frac{\Delta R}{R} = 2\frac{\Delta l}{l} = 2(0.5) = 1$
$21..4.2 = R_0[1 + 0.004(100)]$
$4.2 = R_0[1.4]$

**Level – III**

22. Conceptual
23. Conceptual
24. Use $R = \frac{\rho l}{A}$; $\rho = \frac{RA}{L} = \frac{(0.104\Omega)\pi\left(1.25\times10^{-3}\ m\right)^2}{14}$
$= 3.65 \times 10^{-8}\Omega m$
25. Use V = EL to calculate E& the ohm's law ,
$I = \frac{V}{R} = \frac{EL}{R} = \frac{1.28 \times 14}{0.104} = 172.3\ A$
26. J = I/A (or) J = E/$\rho$ and then use equation for the target variable $V_d$.
$J = n/q/V_d = neV_d$
$V_d = \frac{J}{ne} = \frac{3.51 \times 10^7\ A/m^2}{(8.5 \times 10^{28}\ m^{-3})(1.602 \times 10^{-19}C)}$
$= 2.58 \times 10^{-8}\ m/s = 2.58\ mm/s$
27. Conceptual
28. $1 = 1\ m, r = 0.1\ mm = 0.1 \times 10^{-3}\ m, R = 95.5\Omega; \rho = ?$
$R = \frac{\rho l}{A}$
$\rho = \frac{RA}{1} = \frac{95.5 \times 3.14 \times 0.1 \times 0.1 \times 10^{-6}}{1}$
$= 3 \times 10^{-6}\Omega m$
29. Conceptual
30. Conceptual
31. Specific resistance ($\rho$) is a material property.
It does not depend on 1 (or) A
let
V be the volume of wire, then $V = Al \Rightarrow A = \frac{V}{1}$
Substituting this in $R = \rho\frac{1}{A}$
We have $R = \rho\frac{1^2}{V}$
So, forgiven volume and material(i.e; V and $\rho$arecons tan t )
$R\alpha 1^2$
When 1 is doubled, resistance will become four times or the new resistance will be $4R$.
32. The resistance of the coil is given by $R = \rho\frac{1}{A}$
Where ' L ' is the length of the wire, r is the resistivity of copper and ' A ' is the cross-
sectional area of the wire. Where ' $r$ ' is the radius of the coil, then
$L = (250)2\pi r$
$= (250)(2\pi)(0.12\ m) = 188.5\ m$
If $r_w$ is the radius of the wire itself then its cross-sectional area is
$A = \pi r^2\omega = \pi(0.65 \times 10^{-3}\ m)^2 = 1.33 \times 10^{-6}\ m^2$
According to table $26 - 1$, the resistivity of copper is $\rho = 1.69 \times 10^{-8}\Omega - m$.
Thus,
$R = \frac{\rho L}{A} = \frac{(1.69 \times 10^{-8}\Omega m)(188.5\ m)}{1.33 \times 10^{-6}\ m^2} = 2.4\Omega$
33. We use J = E/$\rho$, Where ' E ' is the magnitude of the (uniform) electric field in the wire. J is the magnitude of the current density and $\rho$ is the resistivity of the material.

The electric field is given by E = V/L, where V is the potential difference along the wire and L is the length of the wire.
Thus,
J = V/L$\rho$ and

$$\rho = \frac{V}{LJ} = \frac{115\ V}{(10\ m)(1.4 \times 10^4\ A/m^2)} = 8.2 \times 10^{-4}\Omega - m$$

## WORKSHEET-3

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. B | 2. C | 3. C | 4. A | 5. A | 6. B | 7. A |
| 8. B | 9. B | 10. D | 11. C | | | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. D | 2. C | 3. A | 4. D | 5. D | 6. D | 7. C |
| 8. B | 9. C | 10. B | 11. B | 12. C | 13. D | 14. A |
| 15. D | 16. D | 17. A | 18. A | 19. B | 20. A | 21. B |

### Level – III

| | | | | |
|---|---|---|---|---|
| 22. A, B, C | 23. A | 24. A | 25. A | 26. A |
| 27. a-1,3,5;b-1,3,5;c-1,5;d-1,4,2 | | 28. 8 | 29. A, B, C, D | |
| 30. A, B, C, D | | 31. C | 32. C | 33. D |

## SOLUTIONS

### Level – II

1. No of combinations $x = 2^n$
2. $\frac{1}{R_1} = \frac{1}{12} + \frac{1}{24} = \frac{3}{24}$
   $\frac{1}{R_1} = \frac{3}{24}$
   $R_1 = \Omega$
   $\frac{1}{R_2} = \frac{1}{12} + \frac{1}{24} = \frac{3}{24}$
   $R_2 = \frac{24}{3} = 8\Omega$
   $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
   $\frac{1}{R} = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$
   $R = \frac{8}{2} = 4\Omega$
3. $\frac{1}{R_1} = \frac{1}{15} + \frac{1}{10} =$
   $\frac{1}{R_1} = \frac{2+3}{30} = \frac{5}{30}$
   $R_1 = 6\Omega$
   $R_2 = 10\Omega$
   $R = R_1 + R_2 = 10 + 6 = 16\Omega$
4. Voltmeter reading
   Voltmeter accross 100Ω so reading is 4 V.
   Ammeter reading

$i = \frac{V}{R_{\text{eff}}}$ $R_{\text{eff}} = 50 + 100 = 150\Omega$

$v = 6\Omega v$

$i = \frac{6}{150} = 0.04$ A

5. $R_{eff} = \frac{R}{n^2}$

$= \frac{10}{10^2} = \frac{10}{100} = 0.1\Omega$

6. Total resistance = R, Resistance of each part

$R_1 = \frac{R}{100}$

$\frac{1}{\mathrm{R}_{eff}} = \frac{1}{2R_1} + \frac{1}{2R_1} + \frac{1}{2R_1} \Rightarrow \frac{3}{2R_1} \Rightarrow \mathrm{R}_{eff} = \frac{2R_1}{3}$

7. Effective resistance $= \frac{R\theta(2\pi-\theta)}{4\pi^2}\left(\theta = \frac{\pi}{3}\right) \Rightarrow = \frac{R\left(\frac{\pi}{3}\right)\left(2\pi-\frac{\pi}{3}\right)}{4\pi^2} \Rightarrow$

$= 18\left(\frac{\pi}{3}\right)\left(\frac{5\pi}{3}\right) = 5\Omega/2$

8. $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{8}{6} + \frac{1}{6} + \frac{1}{12} = \frac{3+4+2}{24} = \frac{9}{24}$

9. $RS_1 = R_1 + R_2 = R + R = 2R$

$RS_2 = R_3 + R_4 = R + R = 2R$

$\frac{1}{R_p} = \frac{1}{Rs_1} + \frac{1}{Rs_2} = \frac{1}{2R} + \frac{1}{2R} = \frac{2}{2R} = \frac{1}{R}$

10. $R^1 = \frac{20}{2} = 5$

11. $R = R_1 + R_2 + \cdots . R_n$

$\int \frac{(n1)}{A} = 1\frac{1}{1} + 2\frac{1}{A} + \cdots n\frac{1}{A}$

$\int n = 1 + 2 + 3 + tn$

$\int n = \frac{n(n+1)}{2}\frac{n+1}{2}s$

12.

$\frac{\mathrm{R}}{5} = 2;\ \mathrm{R} = 10$

$5\mathrm{R} = 50\Omega$

13. Each side has resistance 8Ω The resistance between any two adjacent corners.

$\frac{1}{R} = \frac{1}{24} + \frac{1}{8} = \frac{1+3}{24} = \frac{4}{24}$

$R = \frac{24}{4} = 6\Omega$

14. $2^{n-1} = 2^{3-1}$

$= 2^2$

$= 4$

15. $R_p = \frac{R}{n} pR_s = nR$

$\frac{R_s}{R_p} = \frac{nR}{(R/n)} \Rightarrow \frac{n^2}{1} \Rightarrow R_s = n^2 R_p$

16. $\frac{1}{R_1} = \frac{1}{6} + \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$

$R_1 = 2\Omega$

$\frac{1}{R_2} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$

$\mathrm{R}_2 = 2\Omega$

$\mathrm{R} = R_2 + 2 + 2$

$= 2 + 2 + 2$

$= 6\Omega$

17. $\frac{4}{L_2} = \frac{4}{3}, \frac{r_1}{r_2} = \frac{2}{3}$

$\frac{i_1}{i_2} = \left(\frac{L_2}{L_1}\right)\left(\frac{r_1}{r_2}\right)^2$

$= \left(\frac{3}{4}\right)\left(\frac{2}{3}\right)^2$

$= \frac{1}{3}$

18. $R_s = nR_p$

Each ristor has resistance R.

if no. of resistors 2

then $2R = 2 \cdot \frac{R}{2};\ n = 4$

19. 3,7 are in series we get 10ohm. 10,10 are in parallel we get 5ohm. 5,5 are in series we get 10ohm. 10,10 are in parallel we get 5ohm.

20. 3,6 are in parallel we get 2 ohm. 2,3 are in series we get 5ohm. $i = \frac{V}{R}$

$i = \frac{5}{5} = 1A$. As we already know in parallel connection voltage remains same. Hence combination of 3ohm, 6ohm we get 2ohm. so voltage across 6ohm resistor is equal to 2 volts.

21. wire of length = 100 cm

First part = $1_1$

second part = $100 - 1_1$

If both parts are in parallel connection

$\frac{1_1 \times 100 - 1}{1_1 + 100 - 1_1} = 24$

$1001_1 - 1_2^2 = 2400_1$

$1001 - 1001 - 1_2^2 = 2400 = 0$

$1_1^2 - 1001_1 + 2400 = 0$

$1_1^2 - 601_1 - 401_1 + 2400 = 0$

$1_1^2 - (1_160) - 40(1_1 - 60) = 0$

$1_1 - 40 = 01 = 40$ cm

$1_1 - 60 = 01 = 60$ cm

**Level – III**

22.. Reff = $2\Omega + 1.5\Omega + 5\Omega = 8.5\Omega$

$I = \frac{20}{8.5} = 2.35$ A

(i) $I_B = I_C$ [sum of currents entering into the junction is equal to the sum of currents leaving the junction]

(ii) $I_A = I_B$ [Current will divide at B&C whereas in case of A not]

23. Conceptual

24. All resistors are in parallel connection

Re $q = \frac{12}{4}3\Omega;\ I = \frac{3}{3} = 1A$

25. All the resistors shown in the figure are connected in parallel between the terminals of the battery. The equivalent resistance R between the terminals

$\frac{1}{R} = \frac{1}{6} + \frac{1}{3}$R = 2R

Now, this 2R&4R resistance all in series and this equivalent resistance is $4 + 2 = 6\Omega$

26. Therefore, equivalent resistance of the network = $6\Omega$

Current drawn from the battery is, $i = \frac{\text{net emf}}{\text{net resistance}} = \frac{18}{6} = 3$ A.

27. Use combination of resistors in series and parallel

Find $R_{eq}$ b/w $\underline{a\&b}, \underline{a\&c}, \underline{b\&c}$ Re $q = \gamma + \gamma + \gamma = 3\gamma$
$R_{eq}$ b/w b&c
$R_{eff}\ b/w\ b\ \&c = \gamma$
$R_{eff}\ b/w$ a $\&c = 2\gamma$
(b) $R_{eff}\ b/w\ a\&b$
$R_{eff} = 3\gamma$
$R_{eff}$ b/w a&c
$R_{eff} = \gamma$
(c) $R_{eff}\ b/w\ a\&b = 2\gamma$
$R_{eff}$ b/w b&c $= \gamma$
$R_{eff}\ b/w\ a\&c = \gamma$
$R_{eff}$ b/w a& b
(d) $R_{eff} = \frac{3\gamma\times\gamma}{3\gamma+\gamma} = \frac{3\gamma^2}{4\gamma}$
$R_{eff} = \frac{3}{4}\gamma + 2\gamma = \frac{3\gamma + 8\gamma}{4} = \frac{11\gamma}{4}$
$R_{eff}\ b/w\ a\&c$
$R_{eff}\ b/w\ a\ \&c = 3\gamma$
$R_{eff}\ b/w\ b\&c$
$R_{eff}\ b/w\ b\ \&c = 2\gamma$

28. $\therefore R_1 + R_2 = 8$

29. $R_{eff}$ b/w a& b
$R_{eff} = \gamma + \gamma + \gamma + \gamma = 4\gamma\Omega$
$R_{eff}$ b/w a &c
$R_{eff} = \gamma + \gamma + \gamma + \gamma + \gamma + \gamma + \gamma = 7\gamma\Omega$
$R_{eff}$ b/w a& d
$R_{eff} = 10\gamma$
( above all are in series connection)
$R_{eff}$ b/w a &c
All are in series connection $R_{eff} = 8\gamma\Omega$

30. Reffb $/wX\&Y = 4\Omega$
$\frac{R_1R_2}{R_1R_2} = 1.5\Omega$
$R_1 + R_2 = 8\Omega$
$R_1R_2 = (1.5)(R_1 + R_2)$
$R_1R_2 = (1.5)(8) = 12\Omega$

31. Given resistances are $R_1R_2$
if $R_1 + R_2 = 8\Omega$
$\frac{R_1R_2}{R_1 + R_2} = 1.5\Omega;$ $R_1R_2 = (1.5)(R_1 + R_2)$
$R_1R_2 = (1.5)(8) = 12\Omega;$ $(R_1 - R_2)^2 = (R_1 + R_2)$
$R_1R_2 = (1.5)(8) = 12\Omega$
$(R_1 - R_2)^2 = (R_1 + R_2)^2 - 4R_1R^2$
$= 64 - 48 = 16\ = 64 - 48 = 16$
$R_1 - R_2 = 4\Omega$
As we know $R_1 + R_2 = 8\Omega$, $2R_1 = 12\Omega$ $\therefore \frac{R_1 = 6\Omega}{R_2 = 2\Omega}$

32. Let $R_1\&R_2$ be the individual resistance of the two conductors In series combination, $R_1 + R_2 = \rightarrow (1)$
In parallel combination, $\frac{R_1R_2}{R_1+R_2} = 4 \Rightarrow R_1R_2 = 72 \rightarrow (2)$
$(R_1 - R_2)^2 = (R1 + R_2)^2 - 4R_1R_2 = 36$
$\therefore R_1 - R_2 = 6$
On solving (1) &(3) are get $R_1 = 12\Omega \times R_2 = 6\Omega$

33. $R_{eff} = \frac{R}{n} = \frac{25}{5} = 5\Omega$

# WORKSHEET-4

## Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. C | 2. D | 3. A | 4. A | 5. B | 6. A | 7. A |
| 8. A | 9. A | 10. A | 11. B | 12. A | 13. A | 14. D |
| 15. A, B, C | 16. a-1; b-3; c-4; d-2 | | 17. A, B, D | 18. A, B, C | 19. B | 20. A |
| 21. B | | | | | | |

# SOLUTIONS:

## Level – I

1. $R_{eff} = 4 + 2 = 6\Omega$; $R_{eff} = \frac{6\times3}{6+3} = 2\Omega$
   $R_{eff} = 2\Omega$; $R_{eff} = 6\Omega$
2. $R_{eff} = 12 + 6 = 18\Omega$
   $R_{eff} = \frac{18}{2} = 9\Omega$; $R_{eff} = 9\Omega$
3. $R_{eff}$ when switch is closed.
   $R_{eff} = \frac{6\times12}{6+12} = \frac{72}{18} = 4\Omega$
   $R_{eff} = 4 + 4 = 8\Omega$
4. $R_{eff} = \frac{24}{2} = 12\Omega$ $R_{eff} = \frac{24}{3}$ $R_{eff} = \frac{24}{4} = 6\Omega$
   $R_{eff} = 12 + 8 + 6 = 26\Omega$
5. By using equipotential method
   When resistors are equal &i in parallel connection $R_{eff} = \frac{R}{n}$
6. $Req = 1 + \frac{R}{R+1} = \frac{R+1+R}{R+1}$
7. $R^2 + R = 1 + 2R$; $R^2 - R - 1 = 0$
   $R = \frac{-6 \pm \sqrt{b^2 - 4ac}}{2a}$
   $= \frac{+1 \pm \sqrt{(-1)^2 + 4\cdot1.1}}{2.1} = \frac{1 \pm \sqrt{1+4}}{2}$
   Since $R \neq -ve$, SoR $= \frac{1+\sqrt{5}}{2}$
8. $R_{eff}3\&3 = 3 + 3 = 6\Omega$
   $R_{eff}$ $6\&6 = \frac{6}{2} = 3\Omega$
   $R_{eff}$ $3\&3 = 3 + 3 = 6\Omega$
   $R_{eff}$ $6\&6 = \frac{6}{2} = 3\Omega$
   $R_{eff}$ $3\&3 = 6\Omega$
   $R_{eff}$ $6\&6 = \frac{6}{2} = 3\Omega$
   $R_{eff}$ $3\&3 = 6\Omega$
   $R_{eff} = \frac{6\times3}{6+3} = \frac{18}{9} = 2\Omega$
9. $\rho_{eq} = \rho_1 + \rho_2$
   $x_{eq} = x_1 + x_2$
   $R_{eq} = R_1 + R_2$ ; where $A_1 = A_2 = A$
   $R_{eq} = R_1 + R_2$
   $\frac{\rho x}{A} = \frac{\rho_1 x_1}{A} + \frac{\rho_2 x_2}{A}$
   $\rho x = \rho_1 x_1 + \rho_2 x_2$
   $\rho x = \frac{\rho_1 x_1 + \rho_2 x_2}{x}$ $[\therefore x = x_1 + x_2]$
   $\rho_{eq} = \frac{\rho_1 x_1 + \rho_2 x_2}{x_1 + x_2}$

10. $R_{eff} = \frac{10\times30}{40} = 7.5$
$R_{eff}$ b/w A& B
$R_{eff} = 7.5\Omega$

11. $R_{eff} = \frac{10\times30}{40} = \frac{30}{4} = 7.5\Omega$
$R_{eff}$ b/w A&C
$R_{eff} = 15\Omega$

12. $R_{eff} = 4\Omega + 6\Omega + 8\Omega = 18\Omega$
$R_{eff} = \frac{18\times9}{18+9} = \frac{18\times9}{27} = 6\Omega$
$V_{AB} = 2.7 \times 6 = 16.2$ V

13. $\frac{X.1}{X+1} + 1 + 1 = X$
$\frac{X}{X+1} + 2 = X$
$x^2 - 2x - 2 = 0$
$x = 1 + \sqrt{3} = 2.73\Omega$
Total resistance $= 0.5 + 2.73 = 3.23\Omega$
equivalent resisitance between points $A\&C$ is,

14. Between points $A\&B$ resistors of $4\Omega, 6\Omega\&8\Omega$ resistance are in series 2 these are in parallel to $9\Omega$ resistor.
Equivalent resistance of series combination is
$R_1 = (4 + 6 + 8)ohm = 18\Omega$
If equivalent resistance between $A\&B$ is
$R = 9 \times 18/(9 + 18)$ omh $= 6\Omega$
potential difference $A\&B$ is
V = IR $= 2.7 \times 6$ V $= 16.2$ V
Current through $9\Omega$ resistor $= 16.2/9 = 1.8$ A.
Current through $4\Omega, 6\Omega\&8\Omega$ resistor $= 2.7 - 1.8 = 0.9$ A.

15. Let $ABCDA$ be the wire of resistances $10\Omega$. We have to calculate the resistance of this loop between the diametrically opposite points A&C. The wires ADC and ABC will have resistance $5\Omega$ each. There two are joined in parallel between $A\&C$. The equivalent resistance $R$ between $A\&C$ is therefore, given by
$R = \frac{5\Omega\times5\Omega}{5\Omega+5\Omega} = 2.5\Omega$.

16. $R_{eff} = \frac{\left[\frac{2n}{3}+r\right]}{\left[\frac{2r}{2}+r+r\right]} = \frac{5r}{8}$
$R_{eff} = \frac{r}{3} + r = \frac{4r}{3}$
$R_{eff} = \frac{2r}{2} = r$; $R_{eff} = \frac{r}{4}$

17. $\frac{1}{R} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$ between 0 and B
$R = \frac{4}{3}$
$\frac{1}{R} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$ between 0 and A
$R = \frac{4}{3}\Omega$
$\frac{1}{R} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}\Omega$
$R = 4/3$ between 0 and F

18. $R_{eff}$ of 3R&3R $= \frac{6R\times3R}{6R+3R} = \frac{18R^2}{9R} = 2R$
(i) Total effective resistance $= 2R + R = 3R$
Total current through circuit $= \frac{V}{R}$
$I = \frac{100}{3R}A$

(ii) potential difference across '3R' Resistance is Diagram
Vacross $2R = IR = \frac{100}{3R} \times 2R = \frac{200}{3}$ V.
Since $6R\&3R$ are in parallel In parallel V is same SO P.D across
$3R = \frac{200}{3}$ volt

19. Let Eq resistance of the circuit be R parallel
Re $ff = \frac{\gamma R}{r+\gamma}$
$R = \frac{R\gamma}{R+\gamma} + \gamma \Rightarrow R = \frac{\gamma R + \gamma^2 + R^2}{R+\gamma}$
$R^2 + \gamma R = \gamma R + \gamma^2 + R\gamma$
$R = \gamma R - \gamma^2 = 0;\ R = \frac{1+\sqrt{5}}{2}\gamma$
From this $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

20. In Series $R_{\text{eff}} = nR$
In parallel $R_{\text{eff}} = \frac{R}{n}$
But $nR - \frac{R}{n} = \frac{x}{y}$
$\Rightarrow R\left(n - \frac{1}{n}\right) = \frac{x}{y}$
$\Rightarrow R\left(\frac{n^2-1}{n}\right) = \frac{x}{y}$
$R = \frac{nx}{y(n^2-1)}$

21. $\mathrm{R}_{eff} = \frac{6}{3}\Omega\frac{6}{3} + 6 = \frac{6+18}{3}$
$= \frac{24}{3} = 8\Omega;\ \mathrm{R}_{eff} = \frac{8}{2} = 4\Omega$

## WORKSHEET-5

### Level – I

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. A | 2. A | 3. B | 4. A | 5. D | 6. A | 7. A |
| 8. A | 9. D | 10. C | 11. D | 12. B | 13. B | |

### Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. B | 2. B | 3. A | 4. D | 5. D | 6. B | 7. B |
| 8. B | 9. A | 10. B | 11. A | 12. D | 13. A | 14. D |
| 15. D | 16. B | 17. A | 18. B | 19. B | 20. B | 21. B |

### Level – III

| | | | | | |
|---|---|---|---|---|---|
| 22. A | 23. A | 24. C | 25. A | 26. A | |
| 27. a-p,q; b-p,q; c-r,t; d-s | 28. A | 29. a-q; b-p; c-r; d-s | | 30. A & B | |
| 31. a-p; b-p; c-p; d- | | | | | |

## SOLUTIONS :

### Level – II

1. Effective $\varepsilon_{mfe} = 4(1.6) = 6v$
Total resistance $= 24 + 6 = 30\Omega$
$i = \frac{e}{R} = \frac{6}{30} = 0.2$ A

2. $I = \frac{E}{R+r}$

$I_1 = \frac{E}{2+r} = 1$

$E - r = 2$

$I_2 = \frac{E}{17+r} = 0.25$

$E = (0.25)(17) + 0.25r$

$E - 0.25r = 0.25 \times 17$

$E - 0.2r = 0.25 \times 17$

(2) - (1)

$\Rightarrow 0.74E = 0.25(17 - 2)$

$3E = (15)$

$E = 5v$

3. Total EMF = (1.5)6

= 9v

$i = \frac{9}{24+6} = \frac{9}{30} = 0.3$ A

Potential drop accross load is = 24(0.3) = 7.2 V

4. $e = \frac{w}{q} = \frac{2}{2} = 1v$

5. $E = i_1(R_1 + r)$

E = 1(5 + r) amd E = 0.6(10 + r)

from Eq//ns

$1(5 + r) = 0.6(10 + r)$

$5 + r = 6 + 0.6r$

$r - 0.6r = 6 - 5$

$0.4r = 1$

$r = \frac{1}{0.4} = 2.5\Omega$

6. $E_{eq} = (N - 2m)E$

$E_{eq} = (12 - 2(3))2$

$E_{eq} = (12 - 6)2$

$= 6(2)$

$E_{\text{eq}} = 12$

7. $i_1 = \frac{E}{R_1+r}$

and $i_1 = \frac{12}{16}$

$\frac{12}{16} = \frac{E}{16+r}$

$12(16 + r) = 16E$

$12(16) + 12r = 16E$

$i_2 = \frac{E}{R_2+r}$

and $i_2 = \frac{11}{10}$

$\frac{11}{10} = \frac{E}{10+r}$

$11(10 + r) = 10E$

$11(10) + 11r = 10E$

from Eq//n(1) and (2)

$r = \frac{20}{7}\Omega$

8. $V = E + ir$

9. $i_s = i_p; \frac{nE}{R+nr} = \frac{E}{R+\frac{r}{n}}$

10. $e = 1.08, v = 0.86. R = 3\Omega$

$r = \left(\frac{E-V}{v}\right)R$

$r = \left(\frac{1.08-0.6}{0.6}\right)3 = \left(\frac{0.48}{0.6}\right)3 = 2.4$

11. $i = \frac{E}{R+\frac{r}{n}} = \frac{2}{6+\frac{1}{4}} = \frac{2\times4}{25} = \frac{8}{25}$

12. $mn = 24, mR = nr$

$mn = 24$

$mr = nr$

$\Rightarrow m(3) = n(0.5)$

$\Rightarrow m30 = n5$

n = 6 m

$m(6m) = 24$

$6m^2 = 24$

$m^2 = 4$

$m = 2$

$n = 12$

13. $E_{\text{eff}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$

$E_{\text{eff}} = \frac{1.25(1) + (0.75)(1)}{1+1}$

$= \frac{2}{2} = 1v$

14. $E_1 + E_2 = i_1(r_1 + r_2 + R)$

$E_1 - E_2 = i_2(r_2 + r_1 + R)$

15. $\frac{m}{n} = \frac{1}{30}, \frac{mn(1.5)}{30m+n} = 1.5$

$\frac{m}{n} = \frac{r}{R}; \ \frac{mn(1.5)}{30m+n} = 1.5$

$\frac{m(30m)1.5}{30m+30m} = 1.5$

$m = 2, n = 30 \times 2 = 60$

$m \times n = 2 \times 60 = 120$

16.

$i = \frac{E}{\frac{r}{n}+R}; \ i = \frac{10}{\frac{1}{2}+2} = \frac{20}{5} = 4$

17. n-series m-parallel

$r_{eq} = \frac{nr}{m}$

For parallel $I = I_{\max}$

$R = r_{\text{eff}}$

$3 = \frac{n(0.5)}{m}$

$n = 6m$

and $nm = 24$

6 m · m = 24

$m^2 = 4$

m = 2

$n = 6 \times 2 = 12$

18. $i = \frac{nE}{R+nr}$

$0.5 = \frac{4 \times 1.5}{(R+2)+4\times1} = \frac{6}{R+6}$

$R = 6$

19. $3 = \frac{E_1+E_2}{r_1+r_2+R}$

$1 = \dfrac{E_1 - E_2}{r_1 + r_2 + R}$

From 1 and 2 we get 2:1

20. $r = \left(\frac{E-V}{V}\right) R$

When i = 1 A.

$V = E - ir;\ E = v + ir$

$E = 29 + 1(r)$

when 2 A

$E = 28 + 2r$

from Eq(1) and (2)

$29 + r = 28 + 2r$

$r = 1\Omega$

from eq//n (1)

$E = 29 + (1)(1)$

$E = 29 + 1 = 30$ V

21. $\frac{E}{2+r} = 0.9 \rightarrow (1);\ \frac{E}{7+r} = 0.3 \rightarrow$

$\dfrac{Eq(1)}{Eq(2)} = \dfrac{7+r}{2+r} = 3$

$7 + r = 6 + 3r;\ r = 0.5$

**Level – III**

22. Conceptual
23. Conceptual
24. Current in the resistor,

$i = \dfrac{E_1 r_2 + E_2 r_1}{R(r_1 + r_2) + r_1 r_2} = \dfrac{6 \times 1.5 + 12 \times 3}{9(3 + 1.5) + 3 \times 1.5}$

$= \dfrac{45}{40.5 + 4.5} = 1$ A

25. Battery e.m.f = nE = 1.5 m

Internal resistance of the battery = nr = 0.2 m

Total circuit resistance = R + nr = 40 + 0.2 m

∴ Circuit Current, $I = \frac{nE}{R+nR}$ or $1.5 = \frac{1.5n}{40+0.2n} \therefore n = 50$

26. $I = \frac{2E}{R}; I^1 = \frac{2E}{r} \Rightarrow \frac{R}{r} = 1:1$
27. Conceptual
28. $v = \frac{E}{R+r}$
29. Conceptual
30. $E = 2$ volt , $r = 1.5\Omega, R = 10\Omega$

$n = 2,\ m = n;\ I = \dfrac{nE}{R + \frac{nr}{m}}$

Substituting values, $I = \frac{2\times2}{10\frac{2\times1.5}{2}} A = 0.35A$

Current in Each branch $= \frac{0.35}{2} A = 0.175A$

Potential difference across $R = 0.35 \times 10$ volt = 3.5 volt

31. Potential at point $A, B, C, D$ is zero.

## WORKSHEET-6

**Level – I**

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. C | 2. B | 3. B | 4. C | 5. B | 6. C | 7. A |
| 8. C | 9. D | 10. C | 11. D | 12. B | 13. B | 14. A |
| 15. C | | | | | | |

**Level – II**

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. A | 2. B | 3. C | 4. B | 5. B | 6. B | 7. C |
| 8. D | 9. A | 10. B | 11. B | 12. D | 13. C | 14. B |
| 15. B | 16. B | 17. A | 18. B | 19. B | 20. C | 21. D |

**Level – III**

| | | | | | | |
|---|---|---|---|---|---|---|
| 22. A, B, C | 23. A | 24. A | 25. D | 26. C | 27. A | 28. A,C,D |
| 29. a-p; b-q; c-r; d-s; | 30. C | 31. C | 32. B | | | |

# SOLUTOINS

**Level – II**

1. Current entering the junction is 4 A.
Current leaving at other junctioin must be 4 A.

$$(2+2) = (1+1.3+i)$$
$$4 = 2.3 + i$$
$$i = 4 - 2.3$$
$$i = 1.7\text{amp}$$

2. According to Wheatstone bridge balancing conditions, 8ohm resistance carries zero current. So resistance can be negligible.

$$\text{Re } q = \frac{8\times 8}{8+8} = \frac{64}{16} = 4\text{amp}$$

$3Q = 10\text{ohm};\ R = 3\text{ohm};\ S = 2\text{ohm}$

$P = ?$

$$\frac{P}{Q} = \frac{R}{S};\ \frac{P}{10} = \frac{3}{2}$$

$P = 15\Omega$

4. $x = x\Omega$ left gap

$R = 3\Omega$ Right gap

$l = 40$ cm

x = ?

$$\frac{x}{R} = \frac{l}{100-x}$$

$$\frac{x}{3} = \frac{40}{100-40};\ \frac{x}{3} = \frac{40}{60}$$

$x = 2\Omega$

5. L = 100 cm = 1 m

x = 4x, R = 5x 1 = ?

$$\frac{x}{R} = \frac{l}{100-l};\ \frac{4x}{5x} = \frac{l}{100-l}$$

$400 - 4l = 5l$

$$l = \frac{400}{9} = 44.44 \text{ cm}$$

6. Balancing Condition for Wheatstone bridge is $\frac{P}{Q} = \frac{R}{S}$

$P = 21\Omega, R = 18\Omega, Q = \left(\frac{8x}{8+x} + 3\right)\Omega\ S = 6\Omega$

$$\frac{P}{Q} = \frac{R}{S}$$

$$\frac{21}{\frac{8x}{8+x} + 3} = \frac{18}{6}$$

$$\frac{21(8+x)}{8x + 3(8+x)} = \frac{18}{6}$$

$$\frac{56+7x}{11x+24}=1$$
$56+7x=11x+24$
$4x=32; x=8\Omega$

7. Potential drop is greater at "C".Compared to D.
$\therefore$ D is at higher potential, C is at lower potential.
Current flows from D to C.

8. $R_{eq}=R_1+\left(\frac{R_2R_3}{R_2+R_3}\right)$
$R_1=3.5,\ R_2=3\Omega$
$R_{eq}=3.5\left(\frac{3\times3}{3+3}\right)$
$=3.5+1.5$
$R_{eq}=5\Omega$
$V=iR$
$10=i(5)$
$i=2$ A

9. $x=R;\ R.B=S$
$l=40$ cm
$\frac{x}{RB}=\frac{l}{100-l}$
$\frac{R}{S}=\frac{40}{60}$
$R=\frac{2S}{3};\ S=25\Omega$
$R=\frac{2\times25}{3}=\frac{50}{3}=16.67\Omega$

10. $\frac{P}{Q}=\frac{R}{S};\ \frac{15}{12}=\frac{\left(\frac{10x}{10+x}\right)}{4}$
$2x=10+x$
$x=10\Omega$

11. Apply KVL to loop ABCA
$5-5i-1i-3+6-3i=0$
$8-9i=0$
$9i=8$
$i=8/9A$

12. Let $i_1, i_2$ are the currents in left and right loop respectively.
apply KVL to left loop ABDA
$-6i,-3(i_1+i_2)+15=0$
$-9i_1-3i_2+5=0$
$3i_1+i_2=5$
apply KVL to Right loop CBDC
$-3i_2-3(i_1+i_2)+30=0$
$-3i_1-6i_2+30=0$
$-i_1-2i_2+10=0$
$i_1+2i_2=10$
Solving (1) & (2)
$i_1=0, i_2=5$ A
Current in the branch $BD=i_1+i_2$
$=0+5$
$=5A$

13. $\frac{P}{Q}=\frac{R}{S}$
$K=\frac{500}{5}$
$\frac{Q}{P}=\frac{R}{S}$

$\frac{1}{K} = \frac{505}{S}$

from (1) & (2)

$\frac{s}{500} = \frac{505}{s}$

$S^2 = 505 \times 500$

$S = 502.5\Omega$

14. $\frac{x}{R} = \frac{l}{100-l}$

$\frac{x}{6} = \frac{50}{100-50}$

$x = 6\Omega$

The effective Resistances in the right gap

Re $q = \frac{6 \times 3}{6+3} = \frac{18}{9} = 2\Omega$

$\frac{x}{R} = \frac{l_1}{100-l_1}$

$\frac{6}{2} = \frac{l_1}{100-l_1}$

$300 - 3l_1 = l_1$

$4l_1 = 300$

$l_1 = \frac{300}{4} = 75$ cm

75 cm is the new balancing length

∴ Balancing length increase by 25 cm

15. $\frac{x}{R} = \frac{l}{100-l}$

$\frac{x}{5} = \frac{50}{100-50}$

$x = 5\Omega$

The effective Resistance in the right gap after Connecting 5Ω in series. = 10Ω

$\frac{5}{10} = \frac{l_1}{100-l_1}$

$2l_1 = 100 - l_1$

$3l_1 = 100$

$l_1 = \frac{100}{3}$

$= 33.3$ cm

16. Let $i_1$ is the current in left branch &$i_2$ is the current in right branch apply KVL to left branch

$-500i_1 - (i_1 + i_2)R + 12 = 0$

$-500i_1 - i_1R - i_2R + 12 = 0$

$i_1R + 500i_1 + i_2R = 12$

$-(i_1 + i_2)R + 2 = 0$

$(i_1 + i_2)R = 2$

But $i_2 = 0$

$i_1R = 2$

$(500 + R)i_1 = 12$

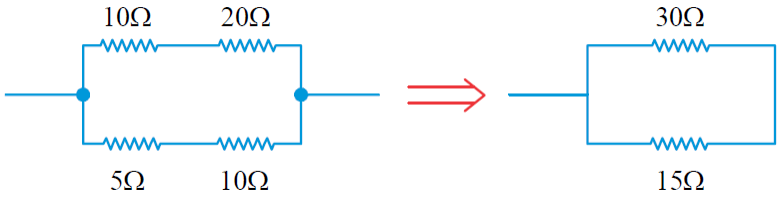
$(500 + r)\left(\frac{2}{R}\right) = 12$

$1000 + 2R = 12R$

$10R = 1000$

$R = 100\Omega$

17.




The circuit forms wheat stone Bridge no current passess through 10Ω Resis tance. The circuit
Reduces to

$$R_{eq} = \frac{30 \times 15}{30 + 15} = \frac{30 \times 15}{45} = 10\Omega$$

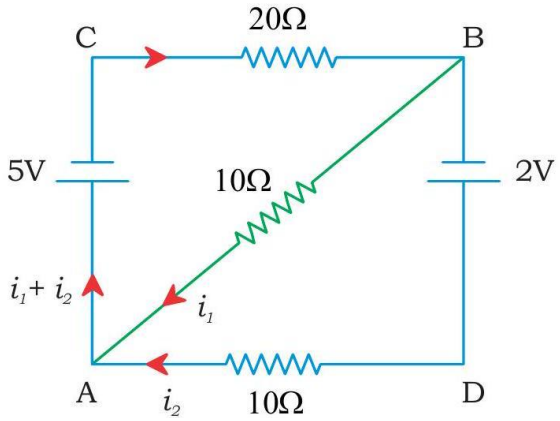
3

$$V = iR$$
$$5 = i(10)$$
$$i = \frac{1}{2}\ A$$
$$= 0.5\ A$$

18. The effective Resistance in left gap $\frac{Xs}{X+S}$

$$\frac{n}{R} = \frac{l}{100 - l} \quad \left(Qn = \frac{Xs}{X + S}\right)$$
$$R = Y$$
$$\frac{\left(\frac{xS}{x+s}\right)}{R} = \frac{l}{100 - l}$$
$$\frac{\left(\frac{5s}{5+s}\right)}{2} = \frac{62.5}{100 - 62.5}$$
$$\frac{5s}{5+s} = 3.34$$
$$5s = (3.34)(5 + s)$$
$$5s = 16.7 + 3.34s$$
$$1.66s = 16.7$$
$$s = 10.06\Omega$$
$$s = 10\Omega$$

19.




Apply KVL to ACB A

$$-5 - 20(i_1 + i_2) - 10i_1 = 0$$
$$-5 - 30i_1 - 20i_2 = 0$$
$$30i_1 + 20i_2 = -5$$
$$6i_1 + 4i_2 = -1$$

Apply KVL to BD AB

$$2 - 10i_2 + 10i_1 = 0$$
$$10i_1 - 10i_2 = -2$$
$$5i_1 - 5i_2 = -1$$

Solving (1) & (2)
we get
$i_1 = \frac{-9}{50}$
$v_A + 10i_1 = v_B$
$v_A - v_B = -10(i_1)$
$= -10\left(\frac{-9}{50}\right)$
$= 1.8$ V

20. $\mathrm{P} = 2\Omega, \mathrm{Q} = 3\Omega, \mathrm{R} = 2\Omega, \mathrm{S} = \frac{4x}{4+x}$
$\frac{P}{Q} = \frac{R}{S}$
$\frac{2}{3} = \frac{2}{(4x/4+x)}$
$\left(\frac{4x}{4+x}\right) = \frac{2\times 3}{2}$
$4x = 3(4+x)$
$4x = 12 + 3x$
$x = 12\Omega$

21. $V_A = 12$ V, $V_D = 6$ V
$V_A - V_D = 6$ V
$R_{eq} = (1+2+3)\Omega = 6\Omega$
$6 = i(6)$
$i = 1A$
$V_A - V_B = iR$
$12 - V_B = 1(1)$
$V_B = 11v$
$V_B - V_c = iR$
$11 - v_c = 1(2)$
$v_c = 11 - 2$
$V_C = 9$ V
$V_C = V_D = iR$
$9 - V_D = 1(3)$
$V_D = 9 - 3$
$= 6$ V

**Level – III**

22. Let $x$ be the potential at the Junction
$V_a - 10i = x\ 30 - 10i = x$
$x - 20i_1 = 12$
$x - 30i_2 = 2$
$(2)\times 3 \Rightarrow 3x - 60i_1 = 36$
$(2)\times 2 \Rightarrow 2x - 60i_2 = 4$
$5x - 60(i_1 + i_2) = 40$
$5x - 60i = 40$
$x - 12i = 8$
$x = 8 + 12i$
sub (4) in (1)
$30 - 10i = 8 + 12i$
$22 = 22i$
$i = 1$ A
$x = 8 + 12(1)$
$x = 20v$
Current in 10Ω Resistor = 1 A

Current in 20Ω Resistor $= \frac{20-12}{20}$

$= \frac{8}{20} = 0.4$ A

Current in 30Ω Resistor $= \frac{20-2}{30}$

$= \frac{18}{30} = 0.6$ A

23. Conceptual [Option - 1]

24. For loop ACEA

$-6(i_1 + i_2) - 1(i_1) + 21 - 5i_1 = 0$

$-6i_1 + 6i_2 - i_1 + 21 - 5i_1 = 0$

$-12\mathrm{i}_1 - 6\mathrm{i}_2 + 21 = 0$

$-4\mathrm{i}_1 - 2\mathrm{i}_2 + 7 = 0$

$4i_1 + 2i_2 = 7$

For loop ACBA

$-6(i_1 + i_2) - 8(i_2 + i_3) + 5 - 4i_2 = 0$

$-6i_1 + 6i_2 - 8i_2 + 8i_3 + 5 - 4i_2 = 0$

$-6\mathrm{i}_1 - 18\mathrm{i}_2 - 8\mathrm{i}_3 + 5 = 0$

$6\mathrm{i}_1 + 18\mathrm{i}_2 + 8\mathrm{i}_3 = 5$

For loop CBDC

$-8(i_2 + i_3) - 16i_3 + 2 = 0$

$-8\mathrm{i}_2 - 8\mathrm{i}_3 - 16\mathrm{i}_3 = -2$

$4i_2 + 4i_3 + 8i_3 = 1$

$4\mathrm{i}_2 + 12\mathrm{i}_3 = 1$

$12\mathrm{i}_3 = 1 - 4\mathrm{i}_2$

$\mathrm{i}_3 = \frac{1 - 4\mathrm{i}_2}{12}$

$6\mathrm{i}_1 + 18\mathrm{i}_2 + 8\mathrm{i}_3 = 5$

$6\mathrm{i}_1 + 18\mathrm{i}_2 + 8\left(\frac{1 - 4\mathrm{i}_2}{\mathrm{i}_2}\right) = 5$

$6i_1 + 18i_2 + \frac{2}{3}(1 - 4i_2) = 5$

$18\mathrm{i}_1 + 54\mathrm{i}_2 + 2 - 8\mathrm{i}_2 = 15$

$18\mathrm{i}_1 + 46\mathrm{i}_2 = 13$

$92i_1 + 46i_2 = 161$

$18i_1 + 46i_2 = 13$

$74\mathrm{i}_1 = 148$

$i_1 = 2$ A

From 1

$4(2) + 2i_2 = 7$

$2\mathrm{i}_2 = -1$

$\mathrm{i}_2 = -0.5$ A

From 3

$4(-0.5) + 12i_3 = 1$

$-2 + 12\mathrm{i}_3 = 1$

$\mathrm{i}_3 = \frac{3}{12}$

$\mathrm{i}_3 = \frac{1}{4} = 0.25$ A

27. Current Entering = Current leaving

$(7 + 1 + 2) = (3 + 2 + 4 + \mathrm{I})$

$10 = 9 + \mathrm{I}$

$i = 1$ A

28. Applying Kirchhoff's first law at the junction B we have $i_1 + i_2 = i_3 \rightarrow (1)$

Applying Kirchoff's Second law to loop ABEFA

$-12 + i_2 \times 1.5 - i_1 \times 1 + 8 = 0$

$i_1 - 1.5i_2 = -4 \rightarrow (2)$
From Loop BCDEB $-(i_2 \times 1.5) - (i_3 \times 9) + 12 = 0$
$1.5i_2 + 9i_3 = 12 \rightarrow (3)$
On solving $i_1 = -1A$ and $i_3 = 1A$

29. Let ' $x$ ' be the current in 2Ω resistor.
$V_A - 4i - 9 - 1(i - x) + 3 - 4i = V_B$
$V_A - V_B = 4i + 9 + 1(i - x) - 3 + 4i$
$16 = 9i + 6 - x \Rightarrow 9i - x = 10$
for the loop
$-2x + 1(i - x) + 9 = 0$
$-2x + i - x + 9 = 0$
$i - 3x = -9$
$i = -9 + 3x$
from (1)
$9(-9 + 3x) - x = 10$
$-81 + 27x - x = 10$
$26x = 91$
$x = \frac{91}{26}$
$x = 3.5$ A (current through 2Ω )
from (1) $9i - x = 10$
$9i - 3.5 = 10$
$9i = 3.5$
$i = 1.5$ A (current through 4Ω )
Current through $1\Omega(i - x) = 1.5 - 3.5$
$= -2$ A
magnitude of current = 2 A
Potential difference across
2Ω resistors $= iR = 3.5(2)$
= 7 V

30. $\frac{x}{40} = \frac{3}{60}$
$x = \frac{3 \times 40}{60}$
$x = 2\Omega$

31. Resistance of wire $= 100 \times 1 = 100\Omega$
effective resistance of left and Right gap $= x + 3$
Total effective Resistance
$= \frac{100 \times (x + 3)}{100 + (x + 3)}$
$= \frac{100 \times 5}{105}$
$= 4.76\Omega$
Current drawn $= \frac{V}{R} = \frac{6}{4.76} = 1.26$ A

32. Current in the upper branch $= \frac{R_2}{R_1 + R_2} \times i$
$= \frac{100}{105} \times 1.26$
$= 1.2$ A

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## WORKSHEET-7

**Level – I**

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. C | 2. A | 3. D | 4. B | 5. B | 6. C | 7. D |
| 8. D | 9. A | 10. A | 11. D | 12. B | | |

**Level – II**

| | | | | | | |
|---|---|---|---|---|---|---|
| 1. B | 2. C | 3. D | 4. C | 5. A | 6. B | 7. B |

| | | | | | | |
|---|---|---|---|---|---|---|
| 8. A | 9. B | 10. A | 11. C | 12. D | 13. C | 14. D |
| 15. A | 16. B | 17. B | 18. B | 19. D | 20. D | 21. C |
| 22.A, B, C | 23. A | 24. A | 25. C | 26. B | 27. a-s; b-p; c-q; d-r | |
| 28. A, D | 29. B, A | 30. C | 31. A, B, C | | | |

**SOLUTIONS:**

**Level – I**

1. $H = \frac{V^2}{R}\Delta t$

$H' = \frac{V^2}{R'}\Delta t$

$R' = \frac{R}{2}$

$H' = \frac{V^2}{\frac{R}{2}}\Delta t$

$H' = 2\left(\frac{V^2}{R}\Delta t\right)$

$H' = 2H$

2. Conceptual

3. $H = \frac{V^2}{R}\Delta t$

$H' = \frac{V^2}{R'}\Delta t$

$H' = \left(\frac{(2V)^2}{R}\Delta t\right)$

$H' = 4H$

4. $H = \frac{V^2}{R}\Delta t;\ H\alpha\frac{1}{R}$

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow Q = Q_1 + Q_2$

thin wire is less than thick wire

5. Conceptual

6. $\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2}$

$P = \frac{P_1P_2}{P_1 + P_2}$

7. $P = P_1 + P_2$

8. Conceptual

9. $H = \frac{V^2}{R}\Delta t$

$H = \frac{i^2R^2}{R}\Delta t$

$H = i^2R\Delta t$

$H = i^2\left(\frac{\rho l}{A}\right)\Delta t$

$H = i^2\left(\frac{\rho l}{\pi r^2}\right)\Delta t$

$H\alpha\frac{l}{r^2}$

$\frac{H_1}{H_2} = \frac{l_1}{l_2} \times \frac{r_2^2}{r_1^2}$

$\frac{H_1}{H_2} = 2$

$= \frac{l}{2l} \times \frac{(2r)^2}{r^2}$

$= \frac{1}{2} \times 4 = 2$

10. Conceptual

11. $H = \frac{V^2}{R}\Delta t$

we know that $R = \frac{\rho l}{A}$

$H = \frac{V^2}{\frac{\rho l}{A}}\Delta t$

$H\alpha\frac{1}{\rho}$

**Level – II**

1. $P = \frac{V^2}{R}$

$25 = \frac{220 \times 220}{R}$

$R = 44 \times 44$

$R = 1936\Omega$

$100 = \frac{220 \times 220}{R}$

$R = 484\Omega$

$P\alpha\frac{1}{R}$

If resistance is more, power will be less then only 25 Watt bulb will fuse.

2. Number of units in a month is KWH then

$= \frac{200 \times 5 + 1500 \times 1}{K} \times$ No. of Days

$= \frac{1000 + 1500}{1000} \times 30$

$= \frac{2500}{1000} \times 30 = 75$ units

3. $P_P = N^2 P_s$

4. $P = \frac{W}{t};\ P = \frac{ms\Delta t}{t}$

$836 = \frac{1 \times 4200(40 - 10)}{t}$

$t = 150\text{sec}$

5. $P = \frac{V^2}{R}$

$600 = \frac{240 \times 240}{R}$

$R = 96\Omega$

6. $P = \frac{P_1 P_2}{P_1 + P_2} = \frac{200 \times 100}{200 + 100}$

$= \frac{20000}{300} = 66\text{Watt}$

7. $P = i^2 R;\ P\alpha i^2$

$\frac{P_1}{P_2} = \frac{i_1^2}{i_2^2}$

$\frac{P_2}{P_1} - 1 = \frac{(110)^2}{(100)^2} - 1$

$\frac{P_2 - P_1}{P_1} = \frac{12100}{10000} - 1$

$\frac{P_2 - P_1}{P_1} \times 100\% = \frac{2100}{10000} \times 100\%$

$= \frac{21}{100} \times 100\% = 21\%$

8. $P = \frac{V^2}{R}$

$P' = \frac{V^2}{R'}$

$R' = \frac{R}{2}$

$P' = \frac{V^2}{\frac{R}{2}}$

$P' = 2P$

9. $KWH \Rightarrow$

$WH = \frac{4\times50\times6+5\times60\times6}{1000} \times 30$ days

$= \frac{1200 + 1800}{1000} \times 30$

$= \frac{3000}{100} \times 3 = 90KWH$

10. $P = \frac{W}{t}$

$210 = \frac{W}{5 \times 60}$

$W = 210 \times 300$

$W = 63000J$

$W = 63000 \times \frac{1}{4.2}$ Cal

$= 15000$ cal

11. $P = \frac{W}{t}$

$i^2R = \frac{10}{t}$

$\Delta t = 1^0C$

12. $P = \frac{W}{t}$

$\frac{V^2}{R} = \frac{ms\Delta t}{t}$

$R = 137$ohm.

13. $p = \frac{w}{t}$

$i^2R = \frac{ms\Delta t}{t}$

$i^2\alpha\Delta t$

14. $p = \frac{W}{t}$

$210 = \frac{w}{5 \times 60}$

$W = 210 \times 300 = 63000$ J

$= 63000 \times \frac{1}{4.2}$ cal $= 15000$cal

15. $H = \frac{V^2}{R}\Delta t$

$H^I = \frac{V^2}{R^1}\Delta t$

$R^1 = \frac{R}{2}.$

$H^1 = \frac{2V^2}{R}\Delta t$

$H^I = 2H$

$\frac{H^1}{H} = \frac{2}{1}$

16. $P = i^2R$

$P\alpha i^2$

$\frac{P_2}{P_1} = \frac{i_2^2}{i_1^2} = \frac{9025}{10000}$

$\frac{P_2}{P_1} - 1 = \frac{9025}{10000} - 1$

-ve indicates decreases.

17. $P_1 = \frac{V^2}{R}$

$R = \frac{R_1 R_2}{R_1 + R_2}, R_1 = \frac{R}{2}, R_2 = \frac{R}{2}.$

finally we get $R = \frac{R}{4}$.

$P_2 = \frac{V^2}{R} = \frac{V^2}{R/4} = 4\frac{V^2}{R} = 4P_1$

18. $P = \frac{V^2}{R}$

$R = \frac{\rho l}{A}$

$P\alpha \frac{1}{l}$

$\frac{P_1}{P_2} = \frac{l_2}{l_1} = \frac{100}{90}$

$\frac{P_2}{P_1} - 1 = \frac{100 - 90}{100} = \frac{10}{90}$

multiply with 100% on both sides we get 11%.

19. kwH ⇒

$\text{wH} = \frac{40 \times 5 \times 5}{1000} \times 30$

$= \frac{1000}{1000} \times 30 = 30$ units

20. Times are connected in parallel

$t_{\text{parallel}} = \frac{t_1 t_2}{t_1 + t_2} = \frac{6 \times 4}{6 + 4} = \frac{24}{10} = 2.4$ min

21. $H = \frac{V^2}{R} \times t, R = \frac{\rho l}{\pi r^2}$

$\frac{l_1}{l_2} = \frac{3}{1}, \frac{\rho_1}{\rho_2} = \frac{1}{20}$

$\frac{R_1}{R_2} = \frac{\rho_1 1_1}{\pi r_1^2} / \frac{\rho_2 1_1}{\pi r_2^2}$

$\frac{R_1}{R_2} = \left(\frac{\rho_1}{\rho_2}\right)\left(\frac{1_1}{1_2}\right)\left(\frac{r_2}{r_1}\right)^2$

$= \left(\frac{1}{20}\right)\left(\frac{3}{1}\right)\left(\frac{2}{1}\right)^2$

$\frac{1}{5} \times 3 = \frac{3}{5}$

$R_1 : R_2 = 3 : 5$

**Level – III**

22. i) $p = \frac{w}{t}$

$\text{p} = \frac{\text{i}^2\text{Rt}}{\text{t}}$

$\text{p} = \text{i}^2\text{R}$

$\begin{pmatrix} \text{V} = \text{iR} \\ \text{R} = \frac{\text{V}}{\text{i}} \end{pmatrix}$

$\text{p} = \text{i}^2 \frac{\text{V}}{\text{i}}$

iii) $P = Vi$
then $W = P.t$
$= Vi.t$
ii) $H = ms\Delta t$

23. Concpetual

24. $U = i^2Rt$
$400 = (2)^2 \times R \times 10$
$400 = 40 \times R$
$R = 10\Omega$

25. $U = i^2Rt = (4)^2 \times 10 \times 10$
$= 16 \times 100 = 1600$ J

26. Concpetual

27. Concpetual

28. $P = \frac{V^2}{R}$
$100 = \frac{220 \times 220}{R}$
$R = 484\Omega$
$100 = 220 \times i$
$i = \frac{10}{22}$
$i = 0.45$ A

29. $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$
$= \frac{(250)^2}{100} = 625\Omega$
resistance $R = \frac{\rho l}{A}$
$= \frac{1.7 \times 10^{-8} \times 10}{5 \times 10^{-6}}$
$= 0.034\Omega$
Total resistance $= 625 + 0.034$
$= 625.034\Omega$
Current $i = \frac{V}{R}$
$= \frac{220}{625.034}$
Power $P = i^2R$
$= \left(\frac{220}{625.034}\right)^2 \times 0.034$
$= (0.35)^2 \times 0.034 = 0.00418$
By the two wires
$= 0.00418 \times 2$
$P = 8.4$ mW

30. $P = \frac{V^2}{R}$
$100 = \frac{220 \times 220}{R}$
$R = 484\Omega$
$P = \frac{V^2}{R}$
$150 = \frac{V^2}{484} \Rightarrow V^2 = 150 \times 484$
$V = \sqrt{150 \times 484} = 269.44 \approx 270$ V

31. Maximum power dissipate only when the internal resistance of the battery will be same $R = r$

$\varepsilon = i(R + r)$
$\varepsilon = 3(r + r)$
$\varepsilon = 3(2r)$
$\varepsilon = 6r$
$P = i^2R$
$9 = (3)^2 \times R$
$9 = 9 \times R$
$R = 1\Omega$
$r = R = 1\Omega$ From $\varepsilon = 6r = 6(1) = 6$ V